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A FAST AND ACCURATE METHOD FOR EVALUATING THE NONRELATIVISTIC FREE-FREE GAUNT FACTOR FOR HYDROGENIC IONS

D. G. HUMMER[1]
Joint Institute for Laboratory Astrophysics, University of Colorado and National Bureau of Standards
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ABSTRACT

A two-dimensional Chebyshev expansion of the nonrelativistic hydrogenic free-free Gaunt factor is presented which can be evaluated extremely rapidly and which gives a maximum relative error of 0.7% over a very wide range of temperatures and frequencies: $-4.0 \leq \log (h\nu/kT) \leq 1.5$ and $-3.0 \leq \log (Z^2\ \mathrm{Ry}/kT) \leq 3$. This expansion is obtained from numerical values computed primarily from the exact expressions of Karzas and Latter, augmented by certain analytic approximations that are valid in regions of the energy plane where the series given by Karzas and Latter become computationally intractable. A brief table of thermally averaged Gaunt factors is given that covers the ranges of $u = h\nu/kT$ and $\gamma^2 = Z^2\ \mathrm{Ry}/kT$ mentioned above in intervals of 0.25 dex.

Subject headings: atomic processes — numerical methods

I. INTRODUCTION

Although the calculation by Karzas and Latter (1961; hereafter KL) of the Gaunt factor for nonrelativistic free-free transitions of a hydrogenic ion has, for many years, provided the primary source for these data, the utility of their work has been limited by the graphical form in which their results are presented. In conjunction with a new, ab initio calculation of opacities for stellar envelopes by the "Opacity Project" (Seaton 1987), the need was felt for a fast and accurate method of generating the free-free absorption coefficient for hydrogenic ions. Such a method would also be very useful in the modeling of stellar atmospheres and winds, where previously a variety of approximations and functional fits to published graphical and tabular material have been used; these procedures are of only moderate accuracy.

We have therefore used the expansion derived by KL to obtain primary values of the free-free Gaunt factor as a function of the initial and final electron energies, augmented by various analytical approximations valid in certain regions of the energy plane for which the expressions given by KL become computationally intractable. The thermal averages are computed numerically, and the resulting values of $\bar{g}_{\mathrm{ff}}(T, \nu)$ are fitted to a two-dimensional Chebyshev expansion in logarithmic temperature and frequency variables. We believe our primary values to be accurate to better than 0.3%. Although we have obtained fits with a maximum error of less than 0.01%, we give here a smaller set of coefficients with an rms fitting error of 0.1% and a maximum fitting error of 0.7%.

II. BASIC EQUATIONS

The cross section for free-free absorption in the field of a nucleus of charge Z involving a transition from E_i to E_f can be written as

$$d\sigma_{\mathrm{ff}}(E_i \rightarrow E_f) = g_{\mathrm{ff}}(\eta_i, \eta_f) a_0^2 Z^2 \frac{32\pi}{3\sqrt{3}} \frac{e^2}{\hbar c} \left(\frac{\mathrm{Ry}}{h\nu}\right)^3 d\left(\frac{h\nu}{\mathrm{Ry}}\right), \tag{2.1}$$

where the cofactor of g_{ff} is the classical cross section (Kramers 1932). The quantities $\eta_{i,f}$ are related to the initial and final energies (for photon absorption) by

$$\eta_{i,f}^2 = Z^2\ \mathrm{Ry}/E_{i,f}\ , \tag{2.2}$$

and the energy $h\nu$ of the absorbed photon by

$$\frac{1}{\eta_f^2} = \frac{1}{\eta_i^2} + \frac{h\nu}{Z^2\ \mathrm{Ry}}\ . \tag{2.3}$$

a) *The Expressions of Karzas and Latter*

KL have expressed the free-free Gaunt factor in terms of series solutions of certain differential equations related to hypergeometric functions. Their result can be written compactly in a form that is advantageous for numerical evaluation,

$$g_{\mathrm{ff}}(\eta_i, \eta_f) = 8\pi\sqrt{3}[\eta_f/(1-\kappa)^2]^2[1 - \exp(-2\pi\eta_i)]^{-1}[\exp(2\pi\eta_f) - 1]^{-1} G_0(\eta_i, \eta_f) \times [(1 + \kappa^2 + 2\eta_f^2) G_0(\eta_i, \eta_f) - \tfrac{4}{3}(\eta_f^2 + \kappa^2)(1 + \eta_f^2) G_1(\eta_i, \eta_f)/(1-\kappa)^2]\ , \tag{2.4}$$

[1] Staff member, Quantum Physics Division, National Bureau of Standards.

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where

$$\kappa = \eta_f/\eta_i < 1 \,. \tag{2.5}$$

The functions $G_l(\eta_i, \eta_l)$, $l = 0, 1$, are expressed as series solutions of ordinary, real differential equations, which have different forms for values of

$$x \equiv -4\eta_i\eta_f/(\eta_i - \eta_f)^2 \tag{2.6}$$

smaller or larger than unity in absolute magnitude. KL expressed these solutions as power series in x or $1/x$ and gave recursion relations for the coefficients. Since, however, both x and the coefficients depend on η_i and η_f, a separate evaluation of the coefficients, followed by multiplication by a power of x or $1/x$, and then summation is not particularly efficient. We have therefore derived recursion relations for the *terms* in the summation rather than the coefficients. Thus, for $|x| < 1$, we have

$$G_l(\eta_i, \eta_f) = \sum_{n=0}^{\infty} a_n(l; \eta_i, \eta_f) \,, \qquad l = 0, 1 \,, \tag{2.7}$$

where

$$a_0 = 1 \,, \qquad a_1 = 2[\eta_f^2 - \kappa(l+1)^2]/[(1-\kappa)^2(l+1)] \,, \tag{2.8}$$

and

$$a_n[n(2l+1+n)] = a_{n-1}\{4\eta_f^2/(1-\kappa)^2 - 4\kappa[(n-1)(2n+4l+1) + (l+1)^2]/(1-\kappa)^2\}$$
$$- a_{n-2}\{4\eta_f^2/(1-\kappa)^2 - 16\kappa^2[(n-2)(n+2l) + (l+1)^2]/(1-\kappa)^4\} \,. \tag{2.9}$$

For $|x| > 1$ we define $y = -1/x$ and write

$$G_l(\eta_i, \eta_f) = y^{l+1}[\cos(\lambda \ln y)A_l(y) + \sin(\lambda \ln y)B_l(y)] \,, \tag{2.10}$$

where

$$\lambda \equiv \tfrac{1}{2}(\eta_i - \eta_f) \,. \tag{2.11}$$

The functions $A_l(y)$ and $B_l(y)$ are expressed as

$$A_l(y) = \sum_{n=0}^{\infty} a_n(l; \eta_i, \eta_f) \,, \qquad B_l(y) = \sum_{n=0}^{\infty} b_n(l; \eta_i, \eta_f) \,, \tag{2.12}$$

where for $l = 0$

$$a_0 = 2C \sin(q_0)/\sqrt{\pi} \,, \tag{2.13}$$

and for $l = 1$

$$a_1 = 12C[(1+\eta_i^2)(1+\eta_f^2)]^{-1/2} \sin(q_1)/\sqrt{\pi} \,; \tag{2.14}$$

for $l = 0$ and 1, b_0 is given by the same expressions as a_0 but with cosine replacing sine. Here

$$C = \{\sinh(\pi\eta_f)\sinh(-\pi\eta_i)/\sinh[\pi(\eta_f - \eta_i)]\eta_i\eta_f(\eta_i - \eta_f)\}^{1/2} \,, \tag{2.15}$$

$$q_0 = \sigma_0(\eta_f - \eta_i) - \sigma_0(\eta_f) + \sigma_0(\eta_i) \,, \tag{2.16}$$

$$q_1 = q_0 - \tan^{-1}\eta_f + \tan^{-1}\eta_i \,, \tag{2.17}$$

and

$$\sigma_0(\eta) \equiv \arg\Gamma(1 + i\eta) \,. \tag{2.18}$$

For $\eta \lesssim 1$, σ_0 is readily evaluated from (Abramowitz 1952)

$$\sigma_0(\eta) = -\eta\gamma_E + \sum_{m=1} (\eta/m - \tan^{-1}\eta/m) \,, \tag{2.19}$$

and for $\eta \gtrsim 1$,

$$\sigma_0(\eta) = \frac{\pi}{4} + \eta(\ln\eta - 1) - \sum_{r=1} \frac{\mathrm{Br}}{[2r(2r-1)\eta^{2r-1}]} \,; \tag{2.20}$$

here γ_E is Euler's constant ($= 0.57721\ldots$) and the coefficients Br are the Bernoulli numbers.

The terms for $n = 1, 2, \ldots$ are generated recursively by

$$a_n = A_n b_n + B_n b_{n-2} + C_n a_{n-1} + D_n a_{n-2} \,, \tag{2.21a}$$

and

$$b_n = -A_n a_{n-1} - B_n a_{n-2} + C_n b_{n-1} + D_n b_{n-2} \,, \tag{2.21b}$$

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where

$$A_n = -\lambda[3n - 2 + 2l(l+1) + 4\lambda^2 + 2\eta_i\eta_f]/\eta_i\eta_f n[4 + (n/\lambda)^2] , \tag{2.22a}$$

$$B_n = -\lambda^3[3n - 4 + 2l(l+1) + 2\lambda^2]/\eta_i^2\eta_f^2 n[4 + (n/\lambda)^2] , \tag{2.22b}$$

$$C_n = \{l(l+1) - (n-1)(2n-1) + \eta_i\eta_f - 6\lambda^2[1 - (1/n)]\}/\eta_i\eta_f[4 + (n/\lambda)^2] , \tag{2.22c}$$

$$D_n = \lambda^2\{l(l+1) - (n-1)(n-2) - 3\lambda^2[1 - (2/n)]\}/\eta_i^2\eta_f^2[4 + (n/\lambda)^2] . \tag{2.22d}$$

We set $a_{-1} = b_{-1} = 0$. The introduction of the ratio κ in equations (2.4) and (2.22) much improves the numerical properties of these relations when $\eta_i \gg 1$. Recursion of terms eliminates certain overflow and underflow problems, but otherwise does not appear to change the stability of the recursion process.

b) *Approximate Forms for* $g_{\rm ff}(\eta_i, \eta_f)$

Although the above equations work well throughout much of that part of the (η_i, η_f)-plane which is of interest here, there are regions in which it becomes computationally intractable and other analytic forms must be used. Both series (2.7) and (2.12) diverge for $x = -1$ and, consequently, converge poorly for x near -1 (or y near 1). This difficulty is easily overcome by quadratic Lagrange interpolation across the "convergence gap," along the curve in the plane generated by the sequence of (η_i, η_f) points used in the particular calculation under consideration.

Although KL do not mention having used approximation formulae for certain limiting cases, we found it necessary to do so. A very useful collection of such formulae and a discussion of their regions of validity is given by Brussard and van de Hulst (1962; hereafter BvdH). An exact limiting case for $\eta_i = \infty$ has been derived by Biedenharn and Thaler (1956) and by Guggenberger (1957) in terms of Coulomb functions, from which one can derive an asymptotic expansion of the form

$$g_{\rm ff}(\eta_i = \infty, \eta_f) \sim 1 + \sum_{n=1} c_n \eta_f^{-2n/3} , \qquad \eta_f > 1 . \tag{2.23a}$$

Guggenberger (1957) gives values of the c_n, $n = 1, 3$; we have calculated the coefficients through $n = 8$ which appear in Table 1. For $\eta_f < 1.7$, BvdH give the expression

$$g_{\rm ff}(\eta_i = \infty, \eta_f) \approx (2\sqrt{3}/\pi)(1 - e^{-2\pi\eta_f}) , \tag{2.23b}$$

which is accurate to better than 1% for $\eta \lesssim 1$. In practice we have used these expressions where $\eta_i \geq 200$ and $\eta_f < 0.1\ \eta_i$.

In the classical limit in which *both* η_i and $\eta_f \gg 1$, a number of authors (see BvdH) have derived an exact limiting form that can be expressed in terms of Hankel functions involving

$$\xi \equiv \eta_i - \eta_f . \tag{2.24}$$

For $\xi > 1$ Grant (1958) derived the asymptotic expansion

$$g^c_{\rm ff}(\eta_i, \eta_f) \sim 1 + 0.21775\xi^{-2/3} - 0.01312\xi^{-4/3} + \cdots , \qquad \xi > 1 , \tag{2.25a}$$

while for $\xi \leq 1$ we have obtained (see Appendix)

$$g^c_{\rm ff}(\eta_i, \eta_f) = (\sqrt{3}/\pi)[\gamma_{\rm E} + \ln(\xi/2)]e^{\pi\xi} {\sum_{n=0}^{n_{\max}}}' a_n T^*_n(\xi) , \qquad 0 \leq \xi \leq 1 . \tag{2.25b}$$

Here T^*_n is the nth-shifted Chebyshev polynomial, and the prime indicates that the first term is halved. The coefficients are given in Table 2 for $n = 0(1)9$; with $n_{\max} = 6$ the maximum relative error is 0.004, while for $n_{\max} = 9$ it is 0.0006. The sum in equation (2.25b) is readily evaluated without calculating the polynomials by using Clenshaw's (1961) summation algorithm. In practice we have used the classical results (2.25) when $\eta_i > 20$ and $\eta_f > 0.1716\eta_i$ (i.e., for $|x| < 1$). Strictly speaking both η_i and η_f should be $\gg 1$, so this lower limit for η_f is really too small. However, a computational procedure using the KL equations breaks down when η_i, η_f, and ξ are all large. This is the weakest part of our calculation and leads to uncertainties of order 1% in the evaluation of $g_{\rm ff}(\eta_i, \eta_f)$ for these values of the parameters.

TABLE 1

COEFFICIENTS c_n FOR ASYMPTOTIC EXPANSION OF $g_{\rm ff}(\eta_i = \infty, \eta_f)$

n	c_n
1	0.17282604
2	−0.04959570
3	−0.01714286
4	0.00204498
5	−0.00243945
6	−0.00120387
7	0.00071814
8	−0.00026971

TABLE 2

COEFFICIENTS a_n FOR THE CLASSICAL APPROXIMATION EQUATION[a]

n	a_n
0	1.47864486
1	−1.72329012
2	0.14420320
3	0.05744888
4	0.01668957
5	0.01580779
6	0.00464268
7	0.00385156
8	0.00116196
9	0.00101967

[a] Eq. (2.25b).

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c) Thermal Average

The thermal average is carried out using the expressions of KL,

$$\bar{g}_{\rm ff}(\gamma^2, u) = \int_0^\infty dt e^{-t} g_{\rm ff}[\eta_i(t), \eta_f(t)] , \tag{2.26}$$

where

$$\eta_i(t) = \gamma/\sqrt{t} , \qquad \eta_f(t) = \gamma/(u + t)^{1/2} \tag{2.27}$$

and

$$u \equiv h\nu/kT , \qquad \gamma^2 \equiv Z^2 \, {\rm Ry}/kT . \tag{2.28}$$

The thermally averaged free-free absorption coefficient is then (cm^{-1} Hz^{-1} sr^{-1})

$$k_\nu^{\rm ff}(T) = N_i N_e \bar{g}_{\rm ff}(T, \nu) \frac{Z^2 e^6}{3\pi h \nu^3 m^2 c} \left(\frac{2\pi m}{3kT}\right)^{1/2} (1 - e^{-h\nu/kT}) . \tag{2.29}$$

The integral was evaluated by repeated application of Simpson's rule using the points $t = 0\ (0.01)\ 0.05\ (0.05)\ 8.0$. This choice was arrived at after a series of convergence tests, which lead us to believe that the relative integration errors are less than 0.001 and that the values of $\bar{g}_{\rm ff}$ are accurate to at least 0.003, again on the relative scale. Although, as mentioned above, some of the values of $g_{\rm ff}(\eta_i, \eta_f)$ are uncertain to within 1%, they enter the integral of t with very small weight; all of the values computed with KL's method are accurate to at least 1 part in 10^4. For the convenience of those who occasionally need values of $\bar{g}_{\rm ff}$, we give in Table 3 a tabulation at the values $\log \gamma^2 = -3.0\ (0.25)\ 3.0$ and $\log u = -4.0\ (0.25)\ 1.5$.

d) Chebyshev Expansion

We have fitted a two-dimensional Chebyshev expansion to our primary data over the intervals $10^{-3} \le \gamma^2 \le 10^3$ and $10^{-4} \le u \le 10^{1.5}$, using the procedure described by Clenshaw (1961). Our final results take the form

$$\bar{g}_{\rm ff}(\gamma^2, u) = \sum_{j=0}^{n}{}' C_j(\gamma^2) T_j(x_u) , \qquad x_u = (2 \log u - 2.5)/5.5 , \tag{2.30}$$

where

$$C_j(\gamma^2) = \sum_{i=0}^{m}{}' D_{i,j} T_i(x_g) , \qquad x_g = (\log \gamma^2)/3 . \tag{2.31}$$

The coefficients $D_{i,j}$ are given in Table 4 for $i = 0, 10$ and $j = 0, 7$, which reproduces the primary data on the grid in u and γ^2 given in Table 3 with an rms fitting error of 0.001 and a maximum fitting error of 0.007. Although slightly larger values of n and m reduced the maximum fitting error by an order of magnitude, it was felt that for practical work the choice $n = 7$ and $m = 10$ is sufficient.

With the Clenshaw summation algorithm on the order of 100 multiplications are required to evaluate the eight coefficients C_n for a given value of γ^2. For each value of u at that value of γ^2, i.e., for each frequency at that temperature, only eight multiplications are required to evaluate $\bar{g}_{\rm ff}$, if a table of values of $\log \nu$ is computed initially and stored. We believe that this method of generating $\bar{g}_{\rm ff}(T, \nu)$ is both considerably more accurate and much faster than any now in use. A FORTRAN subroutine based on this analysis can be obtained from the author; it will also be included among a suite of programs for the evaluation of exact radiative data for hydrogenic ions to be published elsewhere (Storey and Hummer 1988).

III. COMPARISON WITH ANALYTIC APPROXIMATIONS AND OTHER CALCULATIONS

A number of analytic approximations for $\bar{g}_{\rm ff}$ have appeared in the literature; a good discussion is given by BvdH. We consider three of them:

1) A long-wavelength approximation given by Scheuer (1960),

$$\bar{g}_{\rm ff,S}(\gamma^2, u) = -(\sqrt{3}/\pi)[\tfrac{3}{2} \ln \gamma^2 + \ln (u/\gamma^2) - \ln 4 + \tfrac{5}{2}\gamma_{\rm E}] = -0.55133(\tfrac{1}{2} \ln \gamma^2 + \ln u + 0.056745) ; \tag{3.1}$$

2) A high-energy approximation due to Elwert (1954),

$$\bar{g}_{\rm ff,E}(\gamma^2, u) = (\sqrt{3}/\pi)[\ln 4 - \gamma_{\rm E} - \ln u] = 0.55133\ [0.80888 - \ln u] ; \tag{3.2}$$

3) A result based on an asymptotic formula of Menzel and Pekeris (Grant 1958; Seaton 1987),

$$\bar{g}_{\rm ff,MP}(\gamma^2, u) = 1 + 0.1728(u/\gamma^2)^{1/3}(1 + 2u^{-1}) - 0.0496(u/\gamma^2)^{2/3}(1 + \tfrac{2}{3}u^{-1} + \tfrac{4}{3}u^{-2}) . \tag{3.3}$$

We have compared these formulae with our primary data from which Table 3 is drawn, and plotted in Figure 1 the regions in which each approximation gave results within $\pm 10\%$ of our results. For each value of γ^2, the last u value to give agreement within this tolerance was plotted; these points were then connected by straight lines, so the boundaries of these areas are somewhat artificial but the point is clear. Even at this relatively low level of accuracy, the three approximations considered here leave large and important areas of the (u, γ^2)-plane uncovered. At the 1% level, only a tiny fraction of the plane in Figure 1 is covered. Of course, the validity of these formulae extends (and improves) in certain directions beyond the part of the plane considered here.

No significant discrepancies with the graphical presentation of $\bar{g}_{\rm ff}$ given by KL were discovered.

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Recently, Itoh, Nakagawa, and Kohyama (1985) and Nakagawa, Kohyama, and Itoh (1987) have published very extensive calculations of free-free hydrogenic Gaunt factors, which include both relativistic effects in the evaluation of $g_{ff}(\eta_i, \eta_f)$ as well as the effects of electron degeneracy in the evaluation of the thermal average, where the Fermi-Dirac distribution is employed in place of the Maxwellian. Itoh *et al.* give polynomial fits in temperature and frequency to their results for certain values of the chemical potential, and Nakagawa *et al.* provide extensive tables. The case with the weakest degeneracy for which they present data is that with $\eta = -6.0$. For the largest values of γ^2 common to their work and ours, and for the smallest values of u, their values of $\bar{g}_{ff}$ at $\eta = -6.0$ are very slightly smaller than ours. However, it is easily seen by graphical extrapolation of the data of Nakagawa *et al.* to $\eta = -\infty$ that our results agree to at least 0.3%.

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TABLE 3

THERMALLY AVERAGED GAUNT FACTOR $\bar{g}_{ff}(\gamma^2, u)$

A.

	log γ^2												
log u	−3.000	−2.750	−2.500	−2.250	−2.000	−1.750	−1.500	−1.250	−1.000	−0.750	−0.500	−0.250	0.000
−4.00.....	5.516	5.514	5.510	5.504	5.494	5.479	5.456	5.422	5.375	5.311	5.228	5.126	5.007
−3.75.....	5.201	5.199	5.195	5.189	5.179	5.164	5.141	5.108	5.061	4.998	4.915	4.814	4.696
−3.50.....	4.885	4.883	4.880	4.874	4.865	4.850	4.828	4.795	4.749	4.686	4.604	4.504	4.387
−3.25.....	4.570	4.569	4.565	4.560	4.551	4.537	4.515	4.483	4.438	4.376	4.296	4.197	4.082
−3.00.....	4.256	4.254	4.252	4.247	4.239	4.225	4.204	4.173	4.129	4.069	3.990	3.894	3.781
−2.75.....	3.942	3.941	3.939	3.935	3.928	3.915	3.896	3.866	3.824	3.766	3.690	3.596	3.486
−2.50.....	3.630	3.630	3.629	3.626	3.619	3.609	3.591	3.564	3.524	3.468	3.395	3.304	3.198
−2.25.....	3.319	3.320	3.321	3.319	3.315	3.306	3.291	3.266	3.229	3.177	3.108	3.022	2.921
−2.00.....	3.012	3.014	3.016	3.016	3.015	3.009	2.997	2.976	2.943	2.896	2.832	2.751	2.656
−1.75.....	2.708	2.712	2.716	2.719	2.720	2.718	2.710	2.694	2.667	2.626	2.568	2.494	2.407
−1.50.....	2.409	2.415	2.421	2.428	2.433	2.436	2.434	2.424	2.404	2.370	2.320	2.255	2.176
−1.25.....	2.118	2.126	2.135	2.145	2.155	2.164	2.169	2.167	2.155	2.131	2.091	2.036	1.967
−1.00.....	1.837	1.846	1.858	1.872	1.888	1.904	1.917	1.925	1.924	1.911	1.883	1.839	1.783
−0.75.....	1.569	1.580	1.595	1.613	1.634	1.657	1.680	1.700	1.712	1.712	1.698	1.667	1.624
−0.50.....	1.317	1.330	1.347	1.368	1.394	1.425	1.457	1.490	1.517	1.533	1.535	1.521	1.491
−0.25.....	1.087	1.101	1.119	1.143	1.172	1.208	1.250	1.294	1.337	1.372	1.393	1.396	1.382
0.00.....	0.881	0.895	0.914	0.938	0.970	1.009	1.057	1.112	1.170	1.223	1.265	1.290	1.294
0.25.....	0.703	0.716	0.734	0.758	0.789	0.830	0.881	0.942	1.011	1.081	1.145	1.192	1.218
0.50.....	0.552	0.564	0.581	0.603	0.633	0.672	0.723	0.786	0.860	0.943	1.024	1.095	1.145
0.75.....	0.428	0.438	0.453	0.473	0.500	0.536	0.583	0.644	0.719	0.806	0.900	0.991	1.066
1.00.....	0.328	0.337	0.350	0.367	0.390	0.421	0.463	0.519	0.589	0.675	0.774	0.877	0.973
1.25.....	0.250	0.258	0.268	0.282	0.301	0.328	0.363	0.412	0.475	0.554	0.650	0.756	0.865
1.50.....	0.190	0.196	0.204	0.215	0.231	0.252	0.282	0.322	0.376	0.446	0.533	0.636	0.748

B.

	log γ^2											
log u	0.250	0.500	0.750	1.000	1.250	1.500	1.750	2.000	2.250	2.500	2.750	3.000
−4.00.....	4.874	4.732	4.584	4.434	4.282	4.131	3.981	3.831	3.682	3.535	3.390	3.247
−3.75.....	4.564	4.423	4.277	4.128	3.979	3.830	3.682	3.535	3.390	3.247	3.106	2.968
−3.50.....	4.257	4.117	3.973	3.827	3.680	3.534	3.389	3.247	3.106	2.968	2.833	2.701
−3.25.....	3.953	3.816	3.675	3.531	3.388	3.246	3.106	2.968	2.833	2.701	2.572	2.448
−3.00.....	3.655	3.521	3.383	3.243	3.104	2.967	2.833	2.701	2.572	2.448	2.328	2.213
−2.75.....	3.363	3.233	3.099	2.965	2.832	2.700	2.573	2.449	2.329	2.213	2.103	1.998
−2.50.....	3.080	2.955	2.827	2.698	2.572	2.449	2.329	2.214	2.103	1.998	1.899	1.804
−2.25.....	2.808	2.689	2.568	2.447	2.329	2.214	2.104	1.999	1.899	1.806	1.718	1.637
−2.00.....	2.550	2.439	2.325	2.213	2.104	2.000	1.901	1.807	1.719	1.638	1.562	1.494
−1.75.....	2.309	2.206	2.102	2.001	1.902	1.809	1.722	1.640	1.564	1.495	1.432	1.375
−1.50.....	2.088	1.995	1.902	1.811	1.724	1.643	1.567	1.498	1.434	1.377	1.326	1.280
−1.25.....	1.890	1.808	1.726	1.647	1.572	1.502	1.439	1.381	1.329	1.283	1.242	1.206
−1.00.....	1.717	1.647	1.576	1.509	1.446	1.387	1.335	1.288	1.246	1.210	1.178	1.150
−0.75.....	1.570	1.512	1.453	1.397	1.345	1.297	1.254	1.217	1.184	1.155	1.131	1.110
−0.50.....	1.450	1.403	1.355	1.309	1.267	1.228	1.194	1.164	1.138	1.116	1.097	1.081
−0.25.....	1.355	1.320	1.282	1.244	1.210	1.179	1.151	1.127	1.107	1.089	1.073	1.061
0.00.....	1.281	1.257	1.228	1.199	1.171	1.145	1.123	1.102	1.086	1.071	1.058	1.048
0.25.....	1.222	1.211	1.191	1.168	1.145	1.124	1.105	1.089	1.074	1.060	1.049	1.040
0.50.....	1.171	1.176	1.166	1.150	1.131	1.113	1.096	1.081	1.068	1.055	1.044	1.035
0.75.....	1.117	1.142	1.146	1.138	1.124	1.109	1.094	1.080	1.067	1.057	1.043	1.035
1.00.....	1.050	1.100	1.124	1.129	1.121	1.109	1.096	1.083	1.070	1.059	1.048	1.038
1.25.....	0.964	1.041	1.090	1.114	1.118	1.112	1.101	1.088	1.076	1.064	1.055	1.044
1.50.....	0.860	0.959	1.036	1.085	1.107	1.112	1.106	1.096	1.084	1.073	1.062	1.053

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TABLE 4

CHEBYSHEV EXPANSION COEFFICIENT $D_{i,j}$

	J							
I	0	1	2	3	4	5	6	7
0.....	8.986940175+0	−4.009515855+0	8.808871266−1	2.640245111−2	−4.580645915−2	−3.568055702−3	2.827798067−3	3.365860195−4
1.....	−8.006936989−1	9.466021705−1	9.043402532−2	−9.608451450−2	−1.885629865−2	1.050313890−2	2.800889961−3	−1.078209202−3
2.....	−3.781305103−1	1.102726332−1	−1.543619180−2	8.310561114−3	2.179620525−2	4.259726289−3	−4.181588794−3	−1.770208330−3
3.....	1.877213132−2	−1.004885705−1	−5.483366378−2	−4.520154409−3	8.366530426−3	3.700273930−3	6.889320423−4	9.460313195−5
4.....	7.300158392−2	3.576785497−3	−4.545307025−3	−1.017965604−2	−9.530211924−3	−3.450186162−3	1.040482914−3	1.407073544−3
5.....	−1.744671550−3	2.864013856−2	1.903394837−2	7.091074494−3	−9.668371391−4	−2.999107465−3	−1.820642230−3	−3.874082085−4
6.....	−1.707268366−2	−4.694254776−3	1.311691517−3	5.316703136−3	5.178193095−3	2.451228935−3	−2.277321615−5	−8.182359057−4
7.....	2.567331664−4	−9.155339970−3	−6.997479192−3	−3.571518641−3	−2.096101038−4	1.553822487−3	1.509584686−3	6.212627837−4
8.....	4.098322531−3	1.635218463−3	−5.918883504−4	−2.333091048−3	−2.484138313−3	−1.359996060−3	−5.371426147−5	5.553549563−4
9.....	3.837562402−5	2.938325230−3	2.393747064−3	1.328839809−3	9.135013312−5	−7.137252303−4	−7.656848158−4	−3.504683798−4
10.....	−8.491991820−4	−3.615327726−4	3.148015257−4	8.909207650−4	9.869737522−4	6.134671184−4	1.068883394−4	−2.046080100−4

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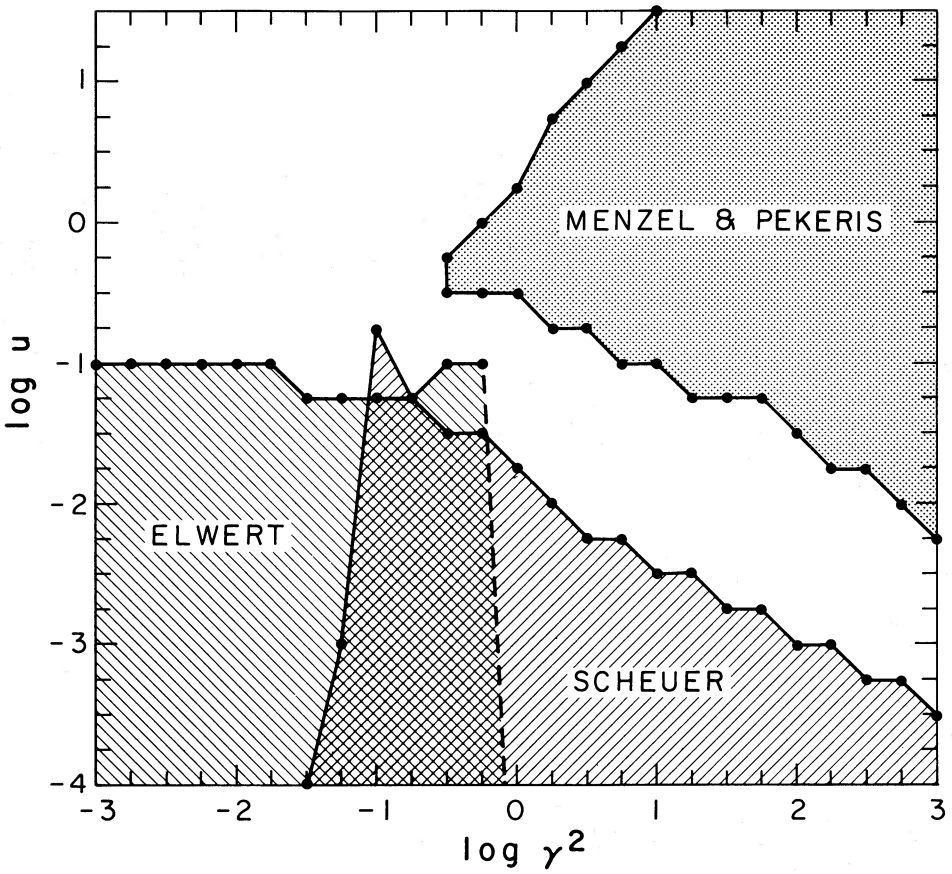


FIG. 1.—Regions of the (u, γ^2)-plane in which the named analytic approximations are within $\pm 10\%$ of the accurate values. Regions are bounded by straight line connecting grid points that lie just within region of agreement. Right boundary of Elwert region is only approximate since its terminus lies beyond log $u = -4.0$. Regions of increasing accuracy lie beyond the boundaries of the figure.

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APPENDIX

Grant (1958) writes the classical expression as (Grant's ξ is defined as the negative of ours)

$$g_{\rm ff}(\xi) = -\left(\frac{\sqrt{3}}{\pi}\right) e^{\pi\xi} \xi K_{i\xi}(\xi) \frac{d}{dz} K_{i\xi}(z)|_{z=\xi} . \tag{A1}$$

From the well-known integral representation of the modified Bessel function of the second kind

$$K_\nu(z) = \int_0^\infty dt e^{-z\cosh(t)} \cosh(\nu t) , \tag{A2}$$

we have

$$K_{i\xi}(\xi) = \int_0^\infty dt e^{-\xi\cosh(t)} \cos(\xi t) , \tag{A3}$$

and

$$\frac{d}{dz} K_{i\xi}(z)|_{z=\xi} = -\int_0^\infty dt e^{-\xi\cosh(t)} \cosh(t) \cos(\xi t) . \tag{A4}$$

Equations (A3) and (A4) are both manifestly real and easy to integrate numerically. The result given in the text (eq. [2.25b]) is obtained by dividing out the nonanalytic part of $K_{i\xi}(\xi)$ and the fitting the results of the numerical integrations using Clenshaw's (1961) procedure.

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D. G. HUMMER: Joint Institute for Laboratory Astrophysics, University of Colorado, Boulder, CO 80309-0440

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